Kinematics Problem Solving

Worked answer key: question, given, diagram, steps with reasons, final answer

Conventions used throughout

• **Notation:** v_1 = initial velocity, v_2 = final velocity, Δd = displacement, Δt = time interval, a = acceleration, $\vec{a}_{av}$ = average acceleration, v_{ave} = average speed. An arrow marks a vector; bars, as in $|\Delta \vec{v}|$, mean its size only. In relative-velocity problems v_{AB} means the velocity of A relative to B.

• **Which object?** When two objects move in one problem, a word in the subscript says which one a symbol belongs to. For example, a_{car} means the car's acceleration.

• **Constant-acceleration equations** (valid only when a is constant): $v_2 = v_1 + a\Delta t$; $\Delta d = \frac{1}{2}(v_1 + v_2)\Delta t$; $\Delta d = v_1\Delta t + \frac{1}{2}a\Delta t^2$; $\Delta d = v_2\Delta t - \frac{1}{2}a\Delta t^2$; $v_2^2 = v_1^2 + 2a\Delta d$.

• **Signs:** each question states which direction is positive. A negative value means the opposite direction. Final answers state the real direction.

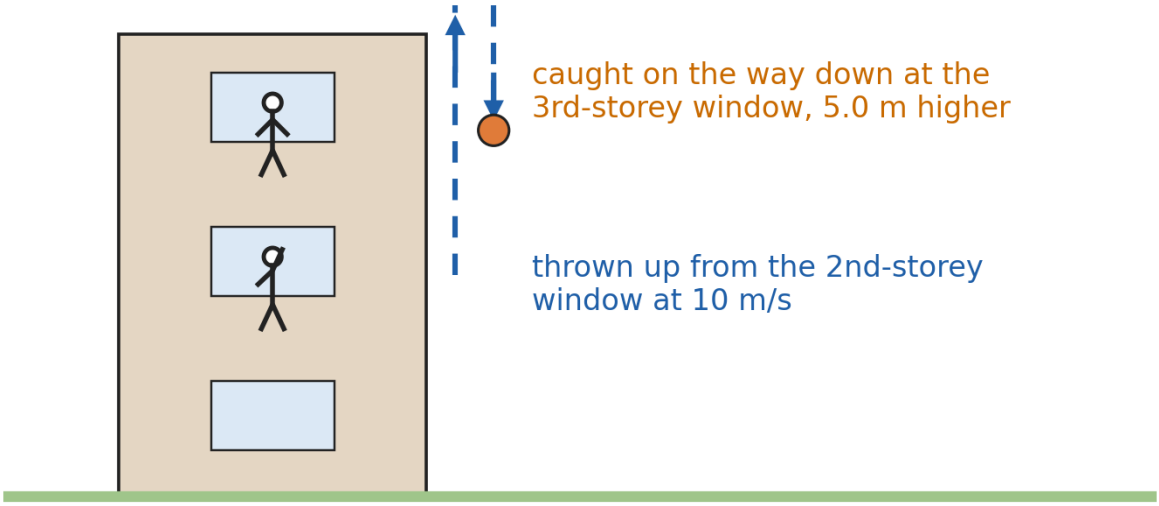
• **$g = 9.81\ m/s^2$** and no air resistance. Full precision is kept in every step; only final answers are rounded.

QUESTION 1

Ball caught on the way down

A ball is thrown vertically upward from a 2nd story window with an initial velocity of 10 m/s. If a person in a third story window that is 5.0 m above the first window catches the ball on its way down, how long was the ball in the air?

Picture the situation



Someone throws a ball straight up out of a window. It rises a little past the window above, stops, and starts falling. A person at the upper window catches it as it comes back down past them.

SOLVING FOR

The time Δt from the throw until the catch.

GIVEN

$\vec{v}_1 = +10$ m/s

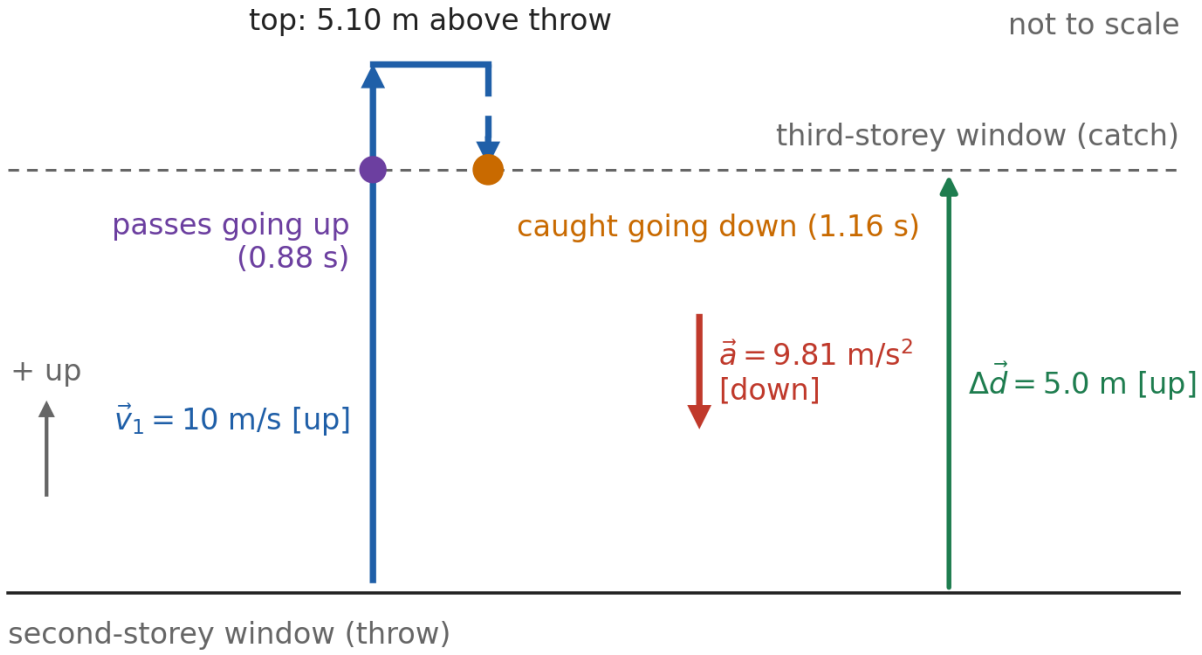
$\Delta\vec{d} = +5.0$ m

$\vec{a} = -9.81$ m/s^2

$\Delta t = ?$

Sign convention: up is positive. The catch is above the throw, so Δd is positive. Gravity points down, so a is negative.

Diagram: motion picture



Step 1 – Choose the equation

The ball moves only under gravity, so its acceleration is constant and the constant-acceleration equations apply. We know v_1, a and Δd, and we want Δt. We don't know (and don't need) the final velocity, so we pick the equation that doesn't contain v_2. Then substitute and move everything to one side.

Formula: $\Delta d = v_1 \Delta t + \frac{1}{2} a \Delta t^2$

Put in the numbers: $5.0 = 10\,\Delta t + \frac{1}{2}(-9.81)\,\Delta t^2$

Simplify $\frac{1}{2}(-9.81) = -4.905$: $5.0 = 10\,\Delta t - 4.905\,\Delta t^2$

Move everything to the left side (add $4.905\Delta t^2$, subtract $10\Delta t$):

$4.905\,\Delta t^2 - 10\,\Delta t + 5.0 = 0$

Step 2 – Solve the quadratic

Δt appears both squared and on its own, so this is a quadratic equation and we can't just isolate Δt. We use the quadratic formula. Match the equation to $a\Delta t^2 + b\Delta t + c = 0$: a = 4.905 (the number in front of Δt^2), b = −10 (in front of Δt) and c = 5.0 (the number on its own).

Formula: $\Delta t = \dfrac{-b \pm \sqrt{b^2 - 4ac}}{2a}$

$$\Delta t = \frac{-(-10) \pm \sqrt{(-10)^2 - 4(4.905)(5.0)}}{2(4.905)}$$

$$\Delta t = \frac{10 \pm \sqrt{100 - 98.1}}{9.81} = \frac{10 \pm \sqrt{1.90}}{9.81} = \frac{10 \pm 1.378}{9.81}$$

With $-$: $\Delta t = \dfrac{8.622}{9.81} = 0.88$ s With $+$: $\Delta t = \dfrac{11.378}{9.81} = 1.16$ s

Step 3 – Pick the right answer

Why two answers? The ball is level with the upper window twice: once on the way up (0.88 s) and once on the way down (1.16 s). The question says it is caught on the way down, so we want the later time. As a check, find the velocity at 1.16 s:

Formula: $v_2 = v_1 + a\Delta t$

$$v_2 = 10 + (-9.81)(1.160) = 10 - 11.38 = -1.38 \text{ m/s}$$

The velocity is negative, so the ball is moving down at that moment. That confirms 1.16 s is the catch.

The ball was in the air for 1.2 s.

The ball only just gets past the upper window. At the top $v_2 = 0$, so $v_2^2 = v_1^2 + 2a\Delta d$ gives a highest point of $10^2/(2 \times 9.81) = 5.10$ m above the throw, only 0.10 m above the catch. That is why the two times are close together.

QUESTION 2

Two opposite vectors

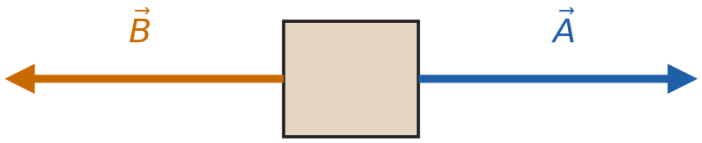
If the sum of two vectors equals zero, the magnitude of their difference is

a) equal to the magnitude of either one

b) equal to twice the magnitude of either one

c) less than twice the magnitude of either one.

Picture the situation



same size, opposite directions: they add to zero

Two arrows that add to zero must cancel exactly, like two equal pulls in opposite directions on a box.

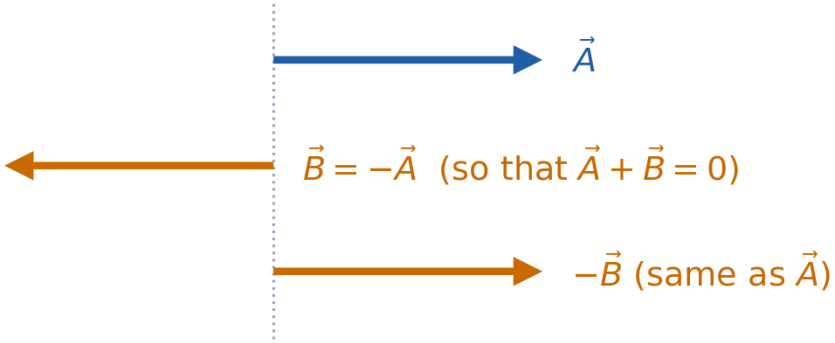
Step 1 – What “the sum is zero” tells us

If $\vec{A} + \vec{B} = 0$, then $\vec{B} = -\vec{A}$. So $\vec{B}$ is the same length as $\vec{A}$ but points the opposite way.

$$\vec{A} + \vec{B} = 0$$

Subtract $\vec{A}$ from both sides: $\vec{B} = -\vec{A}$

Diagram: the two vectors (tails lined up)



Step 2 – Work out the difference

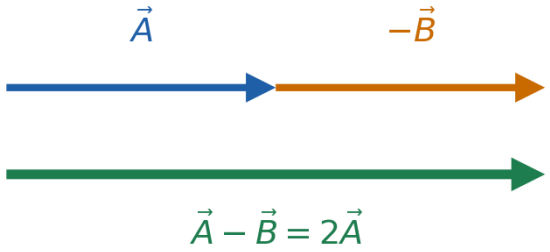
To subtract $\vec{B}$, add its opposite, $-\vec{B}$. But $-\vec{B}$ is the same as $\vec{A}$ (reversing $\vec{B}$ turns it back into $\vec{A}$). So the difference is $\vec{A}$ added to itself: two equal arrows pointing the same way, tip to tail, make one arrow twice as long.

$$\vec{A} - \vec{B} = \vec{A} + (-\vec{B})$$

Since $\vec{B} = -\vec{A}$, then $-\vec{B} = \vec{A}$: $\quad \vec{A} - \vec{B} = \vec{A} + \vec{A} = 2\vec{A}$

So its size is $\quad |\vec{A} - \vec{B}| = 2|\vec{A}|$

Diagram: $\vec{A}$ + $(-\vec{B})$, tip to tail



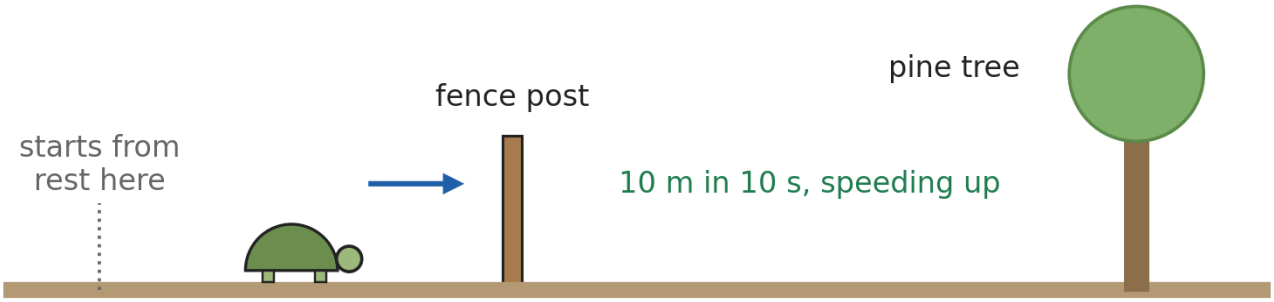
(b) equal to twice the magnitude of either one

QUESTION 3

Turtle: find the rest starting point

A turtle is moving with constant acceleration along a straight ditch. He starts his stopwatch as he passes a fence post and notes that it takes him 10s to reach a pine tree 10m farther along the ditch. As he passes the pine tree his speed is 1.2 m/s. How far was he from the fence post when he started from rest? (SIN 76)

Picture the situation



The turtle started from rest somewhere before the fence post and has been speeding up steadily ever since. We know about the fence-to-tree part and must work backward to where he started.

SOLVING FOR

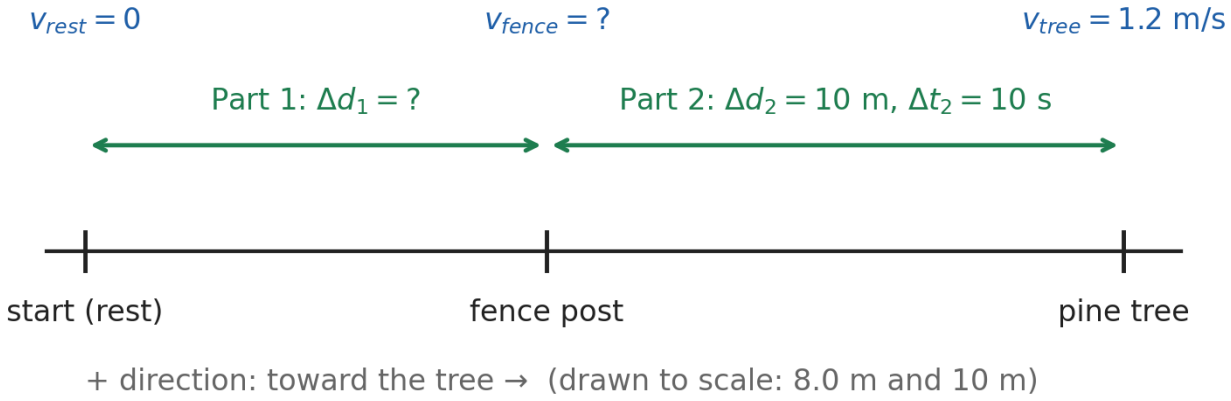
The distance from the starting point (at rest) to the fence post, Δd_1.

GIVEN

$\Delta d_2 = 10$ m	$\Delta t_2 = 10$ s	$v_{tree} = 1.2$ m/s
$v_{rest} = 0$	$a =$ constant	$\Delta d_1 = ?$

Sign convention: + is the direction the turtle moves (toward the tree). He never turns around, so speeds and distances can be used without signs. The trip has two parts. **Part 1** is from the start (at rest) to the fence; **Part 2** is from the fence to the tree. A subscript 1 or 2 on Δd or Δt says which part it belongs to. The speeds are named by place: $v_{rest} = 0$ at the start, v_{fence} at the fence post and v_{tree} at the tree.

Diagram: motion picture



Step 1 – Find his speed at the fence post

We split the trip into two parts and start with the part we know most about: fence to tree. There we know Δd_2, Δt_2 and the final speed. Because the acceleration is constant, the average velocity is halfway between the start and end speeds, so the equation $\Delta d = \frac{1}{2}(v_1 + v_2)\Delta t$ works. For Part 2, v_1 (the speed at the start of the part) is v_{fence} and v_2 (the speed at the end) is v_{tree}. The only unknown is v_{fence}.

Formula: $\Delta d = \dfrac{v_1 + v_2}{2}\,\Delta t$

Multiply both sides by 2: $2\Delta d = (v_1 + v_2)\,\Delta t$

Divide both sides by Δt: $\dfrac{2\Delta d}{\Delta t} = v_1 + v_2$

Subtract v_2 from both sides: $v_1 = \dfrac{2\Delta d}{\Delta t} - v_2$

For Part 2: $v_{fence} = \dfrac{2\Delta d_2}{\Delta t_2} - v_{tree}$

$$v_{fence} = \frac{2(10\text{ m})}{10\text{ s}} - 1.2\text{ m/s} = 2.0 - 1.2 = 0.80\text{ m/s}$$

Step 2 – Find the acceleration

Now we know both speeds on the fence-to-tree part, so the definition of acceleration gives a. This same acceleration applies before the fence too, because it is constant the whole time.

Formula: $a = \frac{v_2 - v_1}{\Delta t}$

For Part 2: $a = \frac{v_{tree} - v_{fence}}{\Delta t_2}$

$a = \frac{1.2 - 0.80}{10} = \frac{0.40}{10} = 0.040 \text{ m/s}^2$

Step 3 – Distance from the start to the fence

Now look at Part 1, from the start to the fence. Here v_1 is $v_{rest} = 0$, v_2 is $v_{fence} = 0.80$ m/s and a = 0.040 m/s². We don't know how long it took, so we use the equation without Δt and solve it for Δd, which here is Δd_1.

Formula: $v_2^2 = v_1^2 + 2a\Delta d$

Subtract v_1^2 from both sides: $v_2^2 - v_1^2 = 2a\Delta d$

Divide both sides by $2a$: $\Delta d = \frac{v_2^2 - v_1^2}{2a}$

For Part 1: $\Delta d_1 = \frac{v_{fence}^2 - v_{rest}^2}{2a}$

$\Delta d_1 = \frac{(0.80)^2 - 0^2}{2(0.040)} = \frac{0.64}{0.080} = 8.0 \text{ m}$

He started from rest 8.0 m before the fence post.

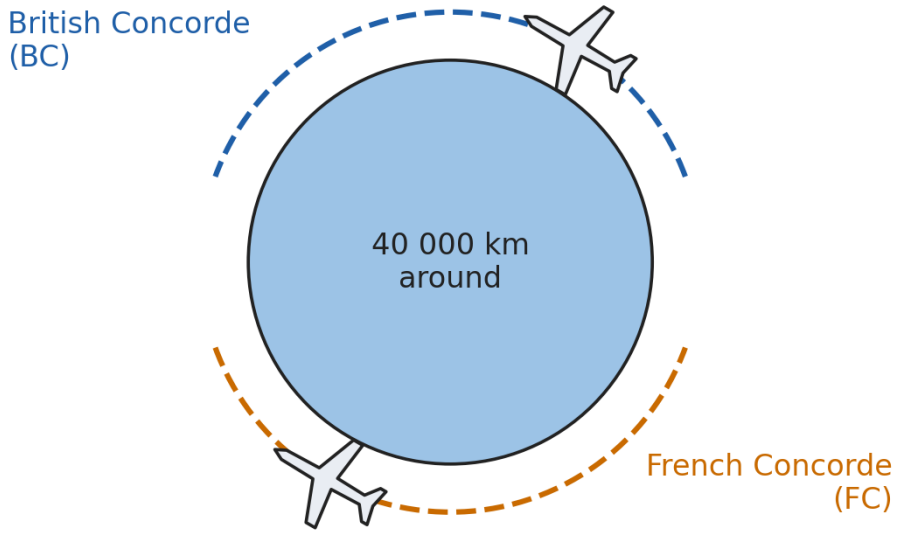
Check: from rest all the way to the tree: $(1.2)^2/(2 \times 0.040) = 18$ m, and 18 m − 10 m = 8.0 m.

QUESTION 4

Concordes: equal distance versus equal time

A British Concorde (BC) and a French Concorde (FC) flew in opposite directions around the earth (40 000 km). The BC covered half of its flight distance at a supersonic speed of 2500 km/h and the other half at a subsonic speed of 1000 km/h. The FC spent half of its flight time at 2500 km/h and the other half at 1000 km/h. Which Concorde completed the trip first, and by how many hours did it beat the other?

Picture the situation



Two planes fly all the way around the earth in opposite directions. They use the same two speeds, but split their trips differently: BC by distance, FC by time.

GIVEN

$d = 40\,000$ km	$v_{fast} = 2500$ km/h	$v_{slow} = 1000$ km/h
$\Delta t_{BC} = ?$	$\Delta t_{FC} = ?$	

Sign convention: only speeds and distances are needed (each plane just goes around once), so no signs are used. d is a distance, v_{fast} and v_{slow} are the two speeds, and Δt_{BC} and Δt_{FC} are the total flight times of the British and French Concordes.

Step 1 – British Concorde (half the distance at each speed)

BC flies 20 000 km fast and 20 000 km slow. For each half, time = distance ÷ speed. Then add the two times.

Formula: $v = \frac{d}{\Delta t}$

Multiply both sides by Δt: $v\,\Delta t = d$

Divide both sides by v: $\Delta t = \frac{d}{v}$

$$\Delta t_{BC} = \frac{20\,000}{2500} + \frac{20\,000}{1000} = 8.0 + 20.0 = 28.0 \text{ h}$$

Step 2 – French Concorde (half the time at each speed)

We don't know FC's total time, Δt_{FC}, yet. It flies for $\Delta t_{FC}/2$ at each speed. The distance in each half is speed × time, and the two distances must add up to 40 000 km. That gives one equation with Δt_{FC} as the only unknown.

Formula: $d = v\,\Delta t$ (for each half)

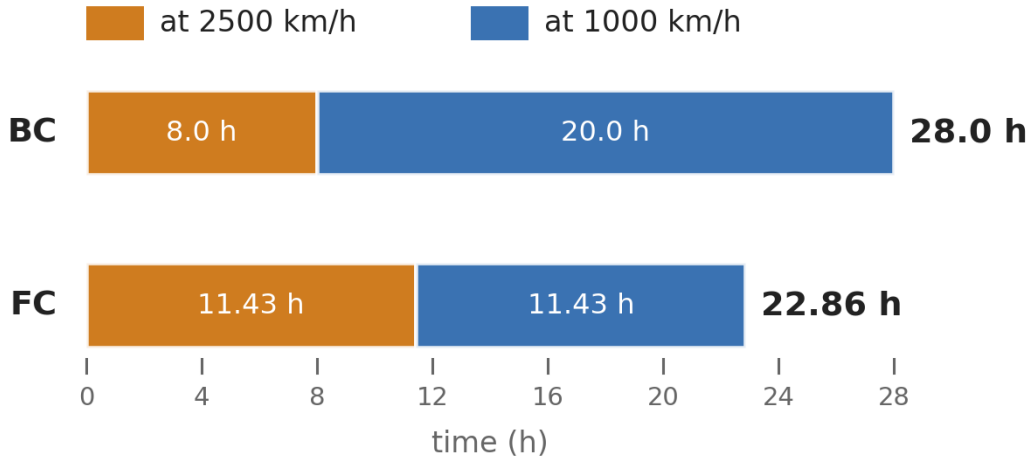
$$2500\left(\frac{\Delta t_{FC}}{2}\right) + 1000\left(\frac{\Delta t_{FC}}{2}\right) = 40\,000$$

Simplify each term: $1250\,\Delta t_{FC} + 500\,\Delta t_{FC} = 40\,000$

Add the like terms: $1750\,\Delta t_{FC} = 40\,000$

Divide both sides by 1750: $\Delta t_{FC} = \frac{40\,000}{1750} = 22.86 \text{ h}$

Diagram: how each plane spends its time (to scale)



Step 3 – Compare

The difference comes from how the time is spent. The slow half of BC's distance takes 20 of its 28 hours, so BC spends most of its time slow. FC spends exactly half its time fast. So FC wins.

$$\Delta t_{BC} - \Delta t_{FC} = 28.0 - 22.86 = 5.14 \text{ h}$$

The French Concorde finishes first, by 5.1 h.

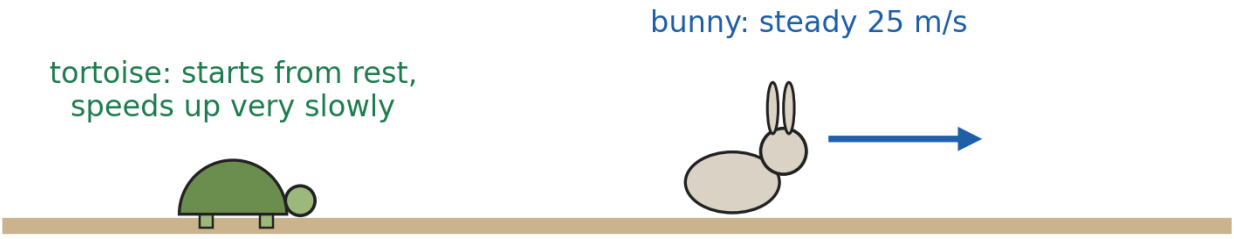
Check: each half lasts $\Delta t_{FC}/2 = 11.43$ h, so FC's distances are $2500 \times 11.43 \approx 28\,570$ km and $1000 \times 11.43 \approx 11\,430$ km, which add to 40 000 km.

QUESTION 5

Bunny and accelerating tortoise

The Easter bunny runs along a straight and narrow path with a constant speed of 25m/s. He passes a sleeping tortoise, which immediately starts to chase the bunny with a constant acceleration of 3.0×10^{-3} m/s^2. How long does it take the tortoise to catch the bunny?

Picture the situation



The bunny zooms past at a steady speed. The tortoise wakes up and starts from rest, speeding up very slowly but steadily. Eventually it catches up.

GIVEN

$v_{bunny} = 25$ m/s

$v_{1(\text{tortoise})} = 0$

$a_{tortoise} = 3.0 \times 10^{-3}$ m/s^2

$\Delta t = ?$

Sign convention: + is the bunny's direction. Both start at the tortoise's position at $\Delta t = 0$. The word in each subscript says which animal a symbol belongs to, and Δt is the time since the bunny passed.

Step 1 – Write how far each one goes in time Δt

The bunny has constant velocity, so its distance is just speed × time. The tortoise starts from rest with constant acceleration, so we use $\Delta d = v_1 \Delta t + \frac{1}{2} a \Delta t^2$ with the tortoise's values: $v_{1(\text{tortoise})} = 0$ and $a_{\text{tortoise}} = 3.0 \times 10^{-3}$ m/s^2.

Formula (bunny, constant velocity): $\Delta d = v\,\Delta t \Rightarrow \Delta d_{bunny} = 25\,\Delta t$

Formula (tortoise): $\Delta d = v_1 \Delta t + \frac{1}{2} a \Delta t^2$

$$\Delta d_{tortoise} = (0)\Delta t + \frac{1}{2}(3.0 \times 10^{-3})\,\Delta t^2 = (1.5 \times 10^{-3})\,\Delta t^2$$

Step 2 – Set them equal

The tortoise catches the bunny when both have gone the same distance from the same start. So we set the two expressions equal. Both sides have a Δt, and we can divide it out because we want the later meeting, not Δt = 0 (the moment they start together).

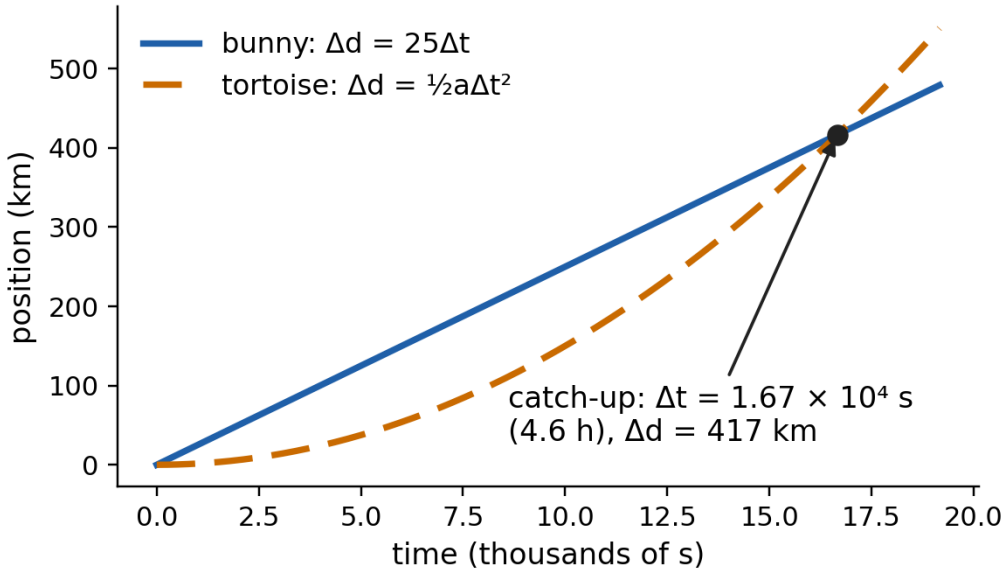
$$25\,\Delta t = (1.5 \times 10^{-3})\,\Delta t^2$$

Divide both sides by Δt: $25 = (1.5 \times 10^{-3})\,\Delta t$

Divide both sides by 1.5×10^{-3}: $\Delta t = \frac{25}{1.5 \times 10^{-3}} = 16\,667$ s

$16\,667 \text{ s} \div 3600 \text{ s/h} = 4.63$ h

Graph: position vs time for both animals



The tortoise catches the bunny after 1.7×10^4 s (about 4.6 h).

Check: at the catch the tortoise is moving at aΔt = 50 m/s, twice the bunny's speed. That makes sense: it started at 0, so to cover the same distance in the same time its average speed must equal the bunny's 25 m/s.

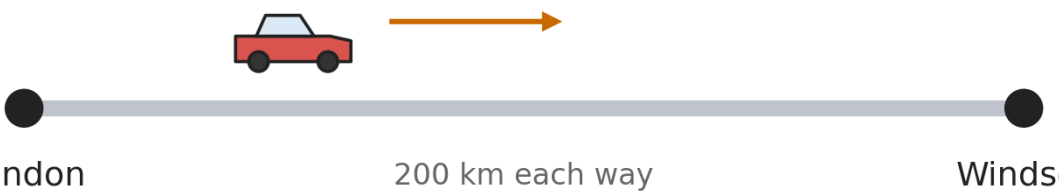
QUESTION 6

Required round-trip average speed

A person travelling between London and Windsor (200 km apart) wants to complete a round trip at an average speed of 100 km/h. On arriving at Windsor at the end of the first leg, she discovers that she has only averaged 75km/h. What must her average speed by on the return trip? (CAP Contest 1984)

Picture the situation

there at 75 km/h, back at ? — average for the round trip must be 100 km/h



She drives 200 km to Windsor and 200 km back. The slow first leg has used up some of her time, so the return leg must be faster.

GIVEN

$d = 200$ km each way $\qquad v_{ave,\,trip} = 100$ km/h

$v_{ave,\,out} = 75$ km/h $\qquad v_{ave,\,back} = ?$

Sign convention: average speed = total distance ÷ total time (speeds, so no directions). “Out” is the first leg (London to Windsor), “back” is the return leg, and “trip” means the whole round trip. So $v_{ave,\,out}$ is her average speed on the first leg and Δt_{back} is the time she has left for the return.

Step 1 – How much time is allowed for the whole trip

The trick is to think in terms of time, not speed. To average 100 km/h over the 400 km round trip, the whole trip must take:

Formula: $v_{ave} = \dfrac{d}{\Delta t}$

Multiply both sides by Δt: $\quad v_{ave}\,\Delta t = d$

Divide both sides by v_{ave}: $\quad \Delta t = \dfrac{d}{v_{ave}}$

$$\Delta t_{trip} = \frac{400 \text{ km}}{100 \text{ km/h}} = 4.00 \text{ h}$$

Step 2 – How much time is left for the return

The first leg used 200 km ÷ 75 km/h of that time. Subtract to find what is left.

$$\Delta t_{out} = \frac{200 \text{ km}}{75 \text{ km/h}} = 2.67 \text{ h}$$

$$\Delta t_{back} = 4.00 - 2.667 = 1.33 \text{ h}$$

Diagram: the 4.00 h time budget



Whole trip must take 400 km ÷ 100 km/h = 4.00 h

Step 3 – Speed needed on the return

She must cover 200 km in the 1.33 h that is left.

Formula: $v_{ave} = \dfrac{d}{\Delta t}$

$$v_{ave,\,back} = \frac{200\ \text{km}}{1.333\ \text{h}} = 150\ \text{km/h}$$

She must average 150 km/h on the return trip.

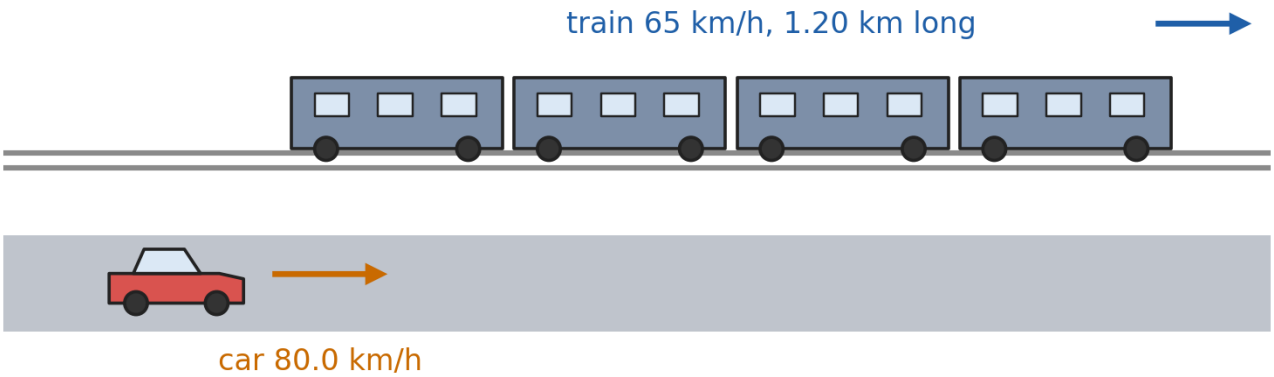
A common wrong answer is 125 km/h (so the two speeds “average” to 100). That fails because she spends more time on the slow leg than on the fast leg, and average speed depends on time, not on the number of legs.

QUESTION 7

Car passing a train

A car going 80.0 km/h overtakes a 1.20 km long train that is travelling at 65 km/h in the same direction on a track parallel to the road. How long does it take the car to pass the train? How long would it take if the vehicles were approaching each other travelling in opposite directions?

Picture the situation



A car on a road beside a railway catches up to a long train and drives past it.

GIVEN

$\vec{v}_{CG} = +80.0$ km/h $\quad |\vec{v}_{TG}| = 65$ km/h $\quad L = 1.20$ km

$\Delta t = ?$ (each case)

Sign convention: + is the car's direction. Subscripts: C = car, T = train, G = ground; v_{CT} is the car's velocity relative to the train. The car's own length is ignored.

Diagram: what "passing" means

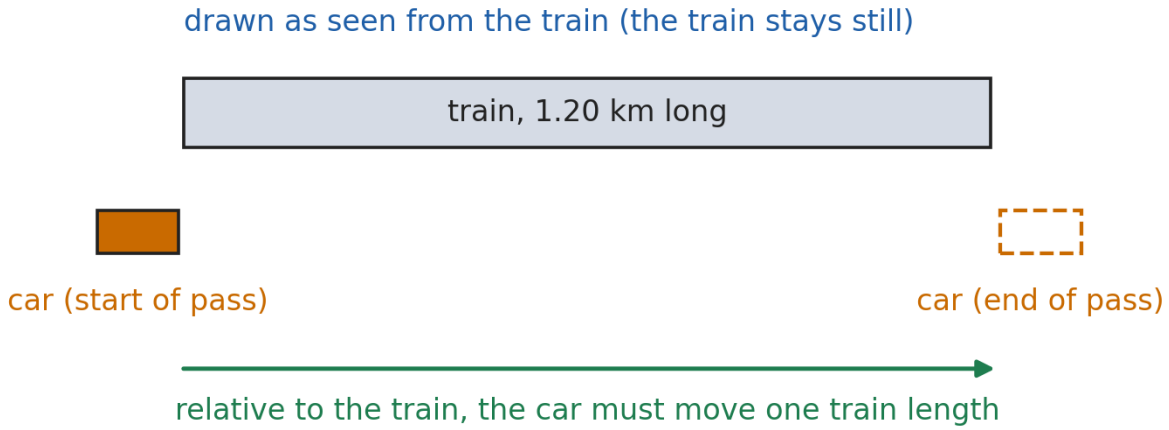
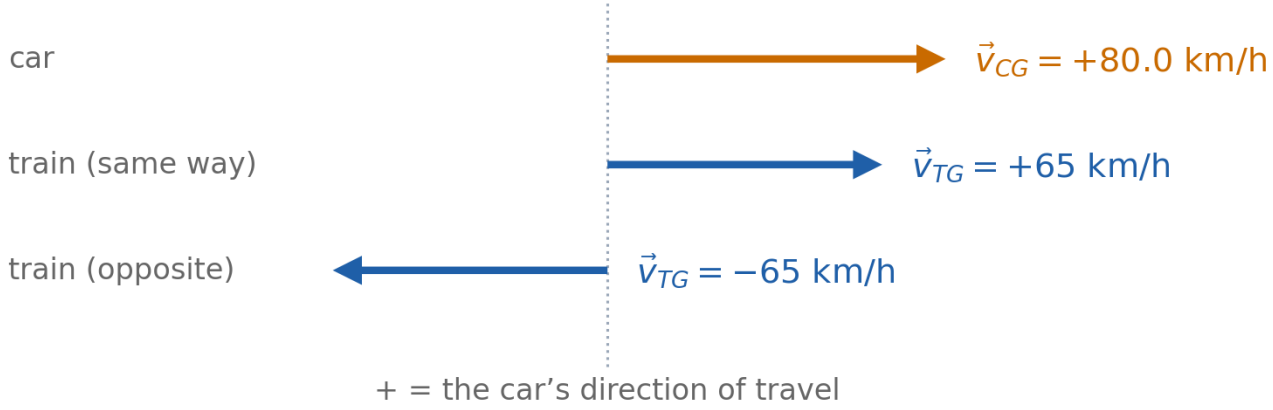


Diagram: velocities relative to the ground



Step 1 – Look at it from the train

The easiest way is to imagine sitting on the train. From there, the train isn't moving, and the car only has to travel one train length, L = 1.20 km, to get from one end of the train to the other. Its velocity as seen from the train, $\vec{v}_{CT}$, is its ground velocity minus the train's ground velocity.

$$\vec{v}_{CT} = \vec{v}_{CG} - \vec{v}_{TG}$$

Step 2 – Same direction

Both move forward, so the car only gains 15 km/h on the train. Time = distance ÷ speed, where the distance is the train length L and the speed is v_{CT}.

$$\vec{v}_{CT} = 80.0 - 65 = +15 \text{ km/h}$$

Formula: $v = \frac{d}{\Delta t}$

Multiply both sides by Δt: $v\,\Delta t = d$

Divide both sides by v: $\Delta t = \frac{d}{v}$

Here $d = L$ and $v = v_{CT}$: $\Delta t = \frac{L}{v_{CT}}$

$$\Delta t = \frac{1.20 \text{ km}}{15 \text{ km/h}} = 0.0800 \text{ h}$$

Convert to minutes (1 h = 60 min): $0.0800 \times 60 = 4.80$ min

Step 3 – Opposite directions

Now the train moves toward the car, so its velocity is −65 km/h. Subtracting a negative adds the speeds: they close the gap much faster.

$$\vec{v}_{CT} = 80.0 - (-65) = 80.0 + 65 = +145 \text{ km/h}$$

Formula: $\Delta t = \frac{L}{v_{CT}}$

$$\Delta t = \frac{1.20 \text{ km}}{145 \text{ km/h}} = 0.008276 \text{ h}$$

Convert (1 h = 3600 s):

$$0.008276 \times 3600 = 29.8 \text{ s} \quad (= 0.497 \text{ min})$$

Same direction: 4.8 min

Opposite directions: 29.8 s (0.497 min)

Significant figures: 80.0 − 65 = 15 has two, so the first answer has two (4.8 min). 80.0 + 65 = 145 has three, so the second has three.